

Multiple Choice (10 marks)

1. Ultraviolet light of 300 nm wavelength strikes the surface of a material which has a work function of 2.0 eV. What is the velocity of the electrons emitted from the surface?
 - (a) 0.87×10^6 m/sec
 - (b) 0.72×10^6 m/sec
 - (c) 0.55×10^6 m/sec
 - (d) 0.40×10^6 m/sec
 - (e) 1.2×10^6 m/sec
2. Which of the following lasers utilizes transitions that involve the energy levels of free atoms?
 - (a) Dye laser
 - (b) Semiconductor laser
 - (c) Gas laser
 - (d) Chemical laser
 - (e) Fiber laser
3. In either a fission event or a fusion event, a quantity that remains unchanged is the
 - (a) kinetic energy
 - (b) number of electrons
 - (c) potential energy
 - (d) temperature of the nucleus
 - (e) mass
4. An ideal battery of constant voltage V supplies current to a circuit that has total resistance R . If R is doubled, what will happen to the total power dissipated by the circuit?
 - (a) It will quadruple.
 - (b) It will increase by a factor of 2.
 - (c) It will become zero.
 - (d) It will be halved.
 - (e) It will double.
5. What is the resistance of a piece of nichrome wire 225 centimeters long with a cross-sectional area of 0.015 square centimeter?

- (a) 15 ohms
- (b) 1.5 ohms
- (c) 2.25 ohms
- (d) 50 ohms
- (e) 0.15 ohms

True / False (10 marks)

Answer True or False

- 6. True or False: When the speed of a moving body increases such that its kinetic energy doubles, its momentum quadruples.
- 7. True or False: The discrete orbits of electrons can be accurately modeled by neutrons, which are not involved in atomic electron behavior.
- 8. True or False: Removing two protons and two neutrons from neon-20 results in the nucleus of magnesium-18.
- 9. True or False: Big brother and little sister can balance on a seesaw due to the equilibrium of forces, torques, and masses acting on the system.
- 10. True or False: The least radiation frequency capable of positron-electron pair production is 7.00×10^{20} Hz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the phenomenon of color mixing in the context of additive color theory, specifically explaining how the overlap of red and blue light results in the color magenta.
- 12. Calculate the horsepower developed by a jet aircraft engine producing a thrust of 3000 lb at a velocity of 600 mi/hr. Discuss the implications of your findings in the context of jet propulsion efficiency.

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